MILITARY SPECIFICATION

MOUNT, TELESCOPE: 9360169

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of mount known as Mount, Telescope: 9360169 used with the US Army mortar systems (see 6.1). Throughout this specification this instrument is referenced as "mount". This mount has radioactively illuminated scales, level vials, and indices.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225	-	Alluminum Alloy Bar, Rod and Wire, Rolled, Drawn or Cold Finished

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of
MIL-L-50778	-	Liquid Scintillation System for Radioactive Fire Control

FSC 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. AMSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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INDUSTRIAL

ASTM B211	-	Alluminum Alloy Bar, Rod, and Wire

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND (AMCCOM)

6523553	-	Elevation and Cross Leveling Fixture
9360169	-	Mount, Telescope
7680631	-	Dioptometer
7691596	-	Azimuth Test Fixture
8658940	-	Inspection Aid
11750708	-	Adapter, Final Inspection, Elevation
11750709	-	Adapter, Final Inspection, Azimuth
11750712	-	Alignment Gage, Final Inspection, Azimuth

PACKAGING DATA SHEETS

9360169	-	Packaging of Mount, Telescope: 9360169

(Copies of drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DSAM 4145.8 - Radioactive Commodites in the DoD Supply System

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)
Part 20: Standards for protection against radiation
Part 30: Rules of general applicability to licensing of by-product material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement possession, handling, and shipping of radioactive material.

3.1.1 External tritium contamination. Tritium in any form shall not be present on the external surfaces of the mount in excess of 1000 dpm/100 sq. cm. subsequent to testing for each of the following requirements: 3.8.1, 3.8.2, 3.8.3, and 3.8.4.

3.1.2 Illumination check. Illuminated areas shall be given a visual check to determine if the light sources are illuminated subsequent to testing for each of the following requirements: 3.8.1, 3.8.2, 3.8.3, and 3.8.4.

3.2 First article. When specified, a sample shall be subjected to First Article Inspection (see 4.2 and 6.2).

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3.3 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.4 Fabrication. The mount shall be manufactured in accordance with Drawing 9360169 and MIL-F-13926.

3.5 General specifications. The following requirements of MIL-F-13926 shall be met:

a. Order of precedence.

b. Dimensions and Tolerances (see ANSI Y14.5).

c. Inorganic protective surface finishes.

d. Parts, identification, and marking.

e. Workmanship.

f. Bonded Joints.

3.6 Temperature conditions. The requirements of this specification shall be met with the mount thermally stabilized at an ambient temperature of between +60°F to +90°F (+16°C to +32°C) unless otherwise specified in the requirement paragraphs.

3.7 Workmanship. The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which affect their function, protective qualities or appearances.

3.8 Environmental. Any mount subjected to any or all of the requirements of 3.8.1, 3.8.2, 3.8.3, and 3.8.4 inclusive, shall subsequently meet the requirements of 3.10 to 3.18 inclusive, unless otherwise specified.

3.8.1 Storage temperature. The mount shall shown no evidence of physical failure such as adhesive separations, glass vial breakage, or casting failure after having been exposed and thermally stabilized at -50°F and +160°F (-46°C and +71°C).

3.8.2 Operating extremes. The mount shall meet the requirements of 3.17 and 3.17.5 inclusive, while exposed to and thermally stabilized at ambient temperatures of -50°F and +145°F (-46°C and +63°C).

3.8.3 Shock. The mount shall be shocked six (6) times in each of three (3) positions. The elevation scale shall be set at the 800 mil mark and the azimuth and elevation locking knobs turned to a locked position with a ten (10) pound-inch torque each. The applied shocks shall be in a vertical, up direction.

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In the first position, the mounting dovetail shall be in the horizontal plane and subjected to six (6) shocks, each at 300g for one-half millisecond. In the second and third positions, the mounting dovetail shall be in a vertical plane with its axis at 45° and 225° and subjected to six (6) shocks each at 200g for one millisecond. The shock pulse shall be a one-half sine wave.

3.8.4 Vibration. The mount shall be vibrated at 30 ± 5 cycles per second frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of five minutes ± 15 seconds. Orientation of the mount shall be in accordance with requirement 3.9. In addition, the azimuth and elevation locking knobs should be turned to a locked position with a ten (10) pound-inch torque each.

3.9 Orientation. The mount shall be positioned as follows to meet the requirements of 3.10 to 3.17.5 inclusive. Where line of sight is specified, the requirement shall be met with the line of sight entering the telescope mounting sleeve from the end opposite to the telescope elevation mechanism.

a. The mounting dovetail surface in a vertical plane.

b. The axis of the dovetail vertical.

c. A line of sight provided which is parallel to the above mounting surface and which is perpendicular to the above axis.

d. The elevation and cross level vial bubbles shall be centered within a half of one graduation space.

e. The coarse azimuth scale index indicating 3200 mils within a half of the width of the graduation line.

f. The coarse elevation scale index indicating 800 mils within a half of the width of the graduation line.

g. The indices for the graduated scales on the azimuth and elevation knobs indicating zero within 0.5 mil.

3.10 Elevation travel. Rotation of the elevation knob shall cause the line of sight to depress a minimum of 800 mils and elevate a minimum of 100 mils from the position specified in 3.9.

3.11 Plumb travel. The line of sight shall track a plumb line within 1.0 mil when elevated and depressed through the range specified in 3.10 and the position specified in 3.9. The total lateral spread shall not exceed 1.0 mil.

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3.12 Lateral displacement. The line of sight shall not exceed 0.5 mil lateral displacement when the elevation motion is reversed by means of the elevation knob.

3.13 Vertical displacement. The line of sight shall not exceed 0.5 mil vertical displacement when the azimuth motion is reversed by means of the azimuth knob.

3.14 Horizontal travel. The line of sight shall not vary in elevation in excess of 1.0 mil (total spread of 2 mils) when rotated 6400 mils in azimuth by means of the azimuth knob.

3.15 Angular accuracy.

3.15.1 Elevation. The angular accuracy with backlash included shall be within 2 mils over the range of elevation travel.

3.15.2 Azimuth. The angular accuracy with backlash included shall be within 2 mils over the range of horizontal travel.

3.16 Backlash.

3.16.1 Elevation. When the amount of backlash at each elevation reading, taken in accordance with 3.15.1, is added to the elevation error at that setting, the sum shall not exceed 2.0 mil.

3.16.2 Azimuth. When the amount of backlash at each azimuth reading, taken in accordance with 3.15.2, is added to the azimuth error at that setting, the sum shall not exceed 2.0 mil.

3.17 Operability.

3.17.1 Telescope clamping mechanism. The telescope clamping mechanism shall be operable in the unlocked and locked positions when using a mandrel test piece having a diameter of .624 - .001 inches. The mandrel shall be fabricated of alluminum alloy 2024 or 6061, T6, QQ-A-225/6 or /8; or ASTM B211. Surface finish 63 minimum. The mandrel shall be annodized. Dimensions apply after annodizing. These requirements shall be met at the ambient temperature of 3.6 and while the mount is subjected to the operating temperatures of 3.8.2.

3.17.1.1 Unlocked position. The mandrel test piece shall rotate freely within its mating diameter and the clamping mechanism shall operate freely about its axis in a fore and aft direction.

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3.17.1.2 Locked position. When locked in place by applying a force of 10 to 12 in-lbs to the wingnut and 15 to 17 in-lbs to the locking lever, there shall be no rotation of the mandrel within the inside diameter of the clamp when a 35 in-lb torque is applied. When locked in place with above forces, the clamping mechanism shall not rotate about its axis when a 15 in-lb torque is applied.

3.17.2 Mount locking latch. The mount locking latch shall be operable at the ambient temperature of 3.6 and while subjected to the operating temperatures of 3.8.2.

3.17.3 Locking knobs. Subsequent to rotating the elevation and azimuth locking knobs to a locked position with a 15 pound-inch torque, there shall be no movement in their respective mechanisms when an 8 pound-inch torque is applied to the elevation and azimuth knobs. This requirement shall be met at the ambient temperature of 3.6 and while the mount is subjected to the operating temperatures of 3.8.2.

3.17.4 Azimuth slip scales. The micrometer and coarse azimuth slip scales shall be operable at the ambient temperature of 3.6 and while subjected to the temperature requirements of 3.8.2.

3.17.5 Torque. The running torque in inch-pounds on all knobs shall be within the limits specified in Table I.

TABLE I. Torque values (inch-pounds)

Knob	+60°F to 90°F (+16°C to +32°C) Standard temperature		-50°F and +145°F (-46°C and +63°C) Extreme operating temperature	
	Min	Max	Min	Max
Elevation	0.5	4.0	0.5	9.0
Azimuth	0.5	4.0	0.5	9.0

3.18 Illumination. The markings of the scales, indices, and level vials should be clearly distinguishable when observed with and without a 50% neutral density filter at light levels of zero to ten foot-candles at a distance of 18 to 20 inches from the viewers eye.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by

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the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts plus three (3) each of all items covered by QAP. The three mounts shall be tested as specified herein for all the requirements of this specification. The items covered by QAP shall be tested, as specified thereon, for all characteristics delineated on the QAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in the production of the balance of the contract quantity.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificated of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Quality conformance inspections.

4.3.1 Submission of product. Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP) listed in the technical data package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Mount. Examination and tests related to Section 3 herein shall be performed on a class basis in accordance with MIL-STD-105, Table II, and the sampling plans specified in Tables II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with DSAM 4145.8 and Section 5 herein. The tabulated clasisification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II. Classification of defects

Class	Characteristic	Requirement	Test procedure
Critical: None			
Major: AQL 0.65% Defective			
101.	Vibration	3.8.4	4.6.3
102.	External tritium contamination	3.1.1	4.6.4
103.	Illumination	3.1.2	Visual
104.	Orientation	3.9.4	4.6.6 & 4.6.7
105.	Elevation travel	3.10	4.6.5.1
106.	Plumb travel	3.11	4.6.5.2
107.	Lateral displacement	3.12	4.6.5.3
108.	Elevation angular accuracy	3.15.1	4.6.5.4
109.	Elevation backlash	3.16.1	4.6.5.5
110.	Horizonal travel	3.14	4.6.6.1
111.	Vertical displacement	3.13	4.6.6.2
112.	Azimuth angular accuracy	3.15.2	4.6.6.3
113.	Azimuth backlash	3.16.2	4.6.6.4
114.	Telescope clamping mechanism, unlocked position	3.17.1.1	4.6.7.1
115.	Telescope clamping mechanism, locked position	3.17.1.2	4.6.7.2
116.	Mount locking latch	3.17.2	4.6.7.3
117.	Locking knobs	3.17.3	4.6.7.4
118.	Azimuth slip scales	3.17.4	4.6.7.5
119.	Elevation knob torque	3.17.5	4.6.7.6
120.	Azimuth knob torque	3.17.5	4.6.7.6
121.	Illumination	3.18	4.6.8
122.	Workmanship	3.7	4.6.9
Minor: None			

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Note: The tests in Table II shall be conducted at room ambient temperature as specified in 3.6.

4.3.3 Noncompliance. The right is reserved to reject any unit of product found defective. Rejected units may be repaired or corrected and resubmitted for inspection with the approval of, and in the manner specified by the procuring activity.

4.4 Quality conformance inspection sampling.

4.4.1 General and environmental. Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall meet the requirements and tests in Table II.

TABLE III. Classification of defects

No.	Characteristic	Requirement	Test procedure
301.	Fabrication	3.4	Applicable Dwg/Visual
302.	General specification	3.5	MIL-F-13926/Visual
303.	Storage/operating temperature	3.8.1	4.6.1
304.	Telescope clamping mechanism, unlocked position	3.8.2, 3.17.1.1	4.6.1, 4.6.7.1
305.	Telescope clamping mechanism, locked position	3.8.2, 3.17.1.2	4.6.1, 4.6.7.2
306.	Mount locking latch	3.8.2, 3.17.2	4.6.1, 4.6.7.3
307.	Locking knobs	3.8.2, 3.17.3	4.6.1, 4.6.7.4
308.	Azimuth slip scales	3.8.2, 3.17.4	4.6.1, 4.6.7.5
309.	Elevation knob torque	3.8.2, 3.17.5	4.6.1, 4.6.7.6
310.	Azimuth knob torque	3.8.2, 3.17.5	4.6.1, 4.6.7.6
311.	Shock	3.8.3	4.6.2
312.	External tritium contamination (post temperature cycle, post shock)	3.1.1	4.6.4
313.	Illumination check (post temperature cycle, post shock)	3.1.2	Visual

Note: The tests for Characteristics 304 through 310 shall be conducted at temperatures of -50°F and +145°F (-46°C and +63°C). The tests for Characteristics 301, 302, and 311 through 313 shall be conducted at room ambient temperature as specified in 3.6.

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4.4.2 Noncompliance of sample. The right is reserved to reject any unit of product found defective. Should any one item of a sampling fail to meet the specified test requirements, acceptance of the represented inspection sample will be suspended by the Government. Rejected samples shall be screened for all defective characteristics. Rejected units may be repaired or corrected and resubmitted for inspection with the approval of, and in the manner specified by the procuring activity.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number in specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list (TDPL).

4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (see 6.3).

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4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall not be so located that radiant heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber in the supply stream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Orientation. The special equipment to position and test the elevation and azimuth accuracy of the mount shall conform to the following:

a. Elevation and Cross Leveling Fixture, 6523553

b. Azimuth Test Fixture, 7691596

c. Adapter, Final Inspection, Elevation, 11750708

d. Adapter, Final Inspection, Azimuth, 11750709

e. Alignment Gage, Final Inspection, Azimuth, 11750712

f. Inspection Aid, 8658940

g. Dioptometer, 7680631

h. Mounting Bracket for Dioptometer

i. Wall Target

4.5.2.3.2.1 Mounting bracket for dioptometer. The mounting bracket shall consist of a sleeve and mandrel joined by a connecting arm. The geometric axis of the sleeve and mandrel shall lay in the same vertical plane within .001 inch. The geometric axes of the sleeve and mandrel shall also be horizontal and parallel to each other within .001 inch. The geometric axis of the sleeve shall be 1.50 + .25 inch above the geometric axis of the mandrel. The inside diameter of the sleeve shall be capable of accepting the dioptometer, 7680631 (outside diameter 1.18 - .01 inch). The sleeve shall secure and maintain the dioptometer in a

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horizontal position. The mandrel shall have a length of 1.000 + .250 inch and an outside diameter of .624 - .001 inch. Install the dioptometer in the sleeve of the bracket. Rotate the dioptometer until its vertical reticle line lays in the same vertical plane that contains the geometric axes of the sleeve and mandrel. Lock the dioptometer in this position.

4.5.2.3.2.2 Wall target. A calibrated wall target shall be utilized in measuring the elevation accuracy requirements of the mount. The wall target shall contain a vertical and horizontal crossline pattern together with limiting displacememt marks needed to determine line of sight deviations (azimuth shift) during elevation travels. This wall target shall be calibrated for a measured distance between 6.0 and 9.0 feet, as measured from the mechanical rotational axis (elevation) of the mount.

4.5.2.3.3 Elevation accuracy. The testing equipment specified in 4.5.2.3.2a, 4.5.2.3.2c, 4.5.2.3.2f, 4.5.2.3.2g, 4.5.2.3.2h, and 4.5.2.3.2i shall be utilized in performing the elevation travel and accuracy tests. The elevation and cross leveling fixture shall be positioned on a vibration free surface and cross leveled in the horizontal plane by means of a certified 10 second level. The Adapter, 11750708, shall be affixed to the elevation and cross leveling fixture and cross leveled by means of the 10 second level so that the interfacing (female) dovetail meets the requirements of 3.9a and 3.9b.

4.5.2.3.4 Azimuth accuracy. The testing equipment specified in 4.5.2.3.2b, 4.5.2.3.2d, 4.5.2.3.2e, 4.5.2.3.2g, 4.5.2.3.2h, and 4.5.2.3.2i shall be utilized in performing the azimuth accuracy tests. The azimuth test fixture shall be positioned on a vibration free surface and cross leveled in the horizontal plane by means of a certified 10 second level. Rotate the wheel assembly of the fixture to the zero azimuth setting. Lock the assembly in this position. Verify that the mechanical axis of the target projector collimator is aligned to the pivotal axis of the wheel assembly of the fixture. The height of the collimator shall be such that its optical axis is coincident to the axis of the sleeve of the mounting bracket when the mount, with mounting bracket and dioptometer affixed, is inserted in the Adapter, 11750709. With the Adapter, 11750709, mounted on the wheel assembly, the Alignment Gage, 11750712, together with the dioptometer, shall align the adapter so that its interfacing (female) dovetail meets the requirement of 3.9a and 3.9b. The alignment gage, with dioptometer, shall also be utilized to establish the mounting dovetail surface of the mount parallel to the optical axis of the collimator.

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4.5.2.3.5 Torque requirements. The torque wrench type tester for testing knobs and clamping mechanism torque shall have a range of zero to fifty inch-pounds. The torque wrench type tester for testing rotation of the mandrel shall have a range of zero to fifty inch-pounds. The accuracy of the torque wrench type testers shall be five percent throughout its entire zone of excursion. The testing equipment shall be equipped with Government approved adapters to accommodate testing of the appropriate assembly.

4.5.2.3.6 Fabrication, setup, and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instructions shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures.

4.6.1 Storage/operating temperature. The testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The mount shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature (as specified in 3.6). The mount shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure, and while at -50°F (-46°C), the mount shall be subjected to the operability tests in 4.6.7.1 through 4.6.7.6. Upon completion of the low temperature operability tests, the temperature of the chamber shall be raised, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature (as specified in 3.6) and the mount allowed to stabilize at this temperature for 2 hours plus or minus 15 minutes, at which time the mount will be subjected to the test of 4.6.7.5. Upon completion of the room ambient temperature (as specified in 3.6) test, the temperature of the chamber shall be raised, over a period of 4 hours plus or minus 15 minutes, to +160°F (+71°C). The mount shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure, the temperature of the chamber shall be reduced, over a period of 30 minutes plus or minus 2 minutes, to +145°F (+63°C). The mount shall be maintained at this extreme temperature for a minimum of 2 hours. Upon completion of the 2 hour exposure and while at +145°F (+63°), the mount shall again be subjected to the operability tests in 4.6.7.1 through 4.6.7.6. Upon completion of the high temperature operability tests, the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature (as specified in 3.6). The mount shall then be removed from the test chamber and subjected to the test in 4.6.4 and then a visual examination for illumination of light sources.

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It shall meet the requirements of 3.1.1 and 3.1.2 respectively subsequent to the temperature exposures of 3.8.2. The mount shall have conformed to the requirements of 3.8.2 at each temperature extreme.

Note: When chamber temperatures are being raised or reduced, the rate of temperature change shall not exceed +2°F (1.1°C) per minute.

4.6.2 Shock. Conduct the shock test with the mount positioned on an adapter affixed to a shock test machine. The adapter shall be capable of holding and positioning the mount as specified in 3.8.3. The number of shocks and applied directions shall be as specified in 3.8.3. The shock pulse shape, peak values, and normal time durations for the positions indicated shall be as specified in 3.8.3. Position the mount (scale and knobs) as specified prior to shock testing. Subsequent to the shock applications (18), the mount shall be removed from the shock test adapter and subjected to the tests in 4.6.4 and then a visual examination for illumination of the light sources. It shall meet the requirements of 3.1.1 and 3.1.2 respectively. The mount shall be subjected to a visual, tactile, and audible examination. It shall show no evidence of physical damage. The mount shall then be subjected to the tests in 4.6.5 through 4.6.8 to determine overall conformance to the requirements of 3.8.3.

4.6.3 Vibration. Conduct the vibration test with the mount positioned on an adapter affixed to a standard vibrating machine that is capable of providing the amplitude and frequency specified in 3.8.4. The adapter shall position the mount as specified in 3.8.4 during the vibration cycle. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle, the mount shall be removed from the vibration adapter and subjected to the test in 4.6.4 and then a visual examination for illumination of the light sources. It shall meet the requirements of 3.1.1 and 3.1.2 respectively. The mount shall be subjected to a visual, tactile, and audible examination. There shall be no evidence of physical damage. The mount shall then be subjected to the tests in 4.6.5 through 4.6.8 to determine overall conformance to the requirements of 3.8.4.

4.6.4 External tritium contamination. The contamination test shall be performed by wiping the item under test with filter paper moistened in distilled water. The damp filter paper shall be placed in a container with the proper percentage of scintillation liquid, which is determined by the type of scintillation test equipment used to determine the actual contamination. The scintillating system used to measure contamination must be calibrated and meet the minimum requirement of MIL-L-50778.

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Insert the sample into this system, count a minimum of 2000 disintergrations and convert the readout into microcuries (222 DPM 10^4=1 microcurie). The measured contamination shall be within the limits specified in 3.1.1. The liquid scintillating system must be calibrated, on six month intervals, to a standard traceable to an N.B.S. standard.

4.6.5 Orientation, elevation accuracy. The testing equipment utilized for these tests shall conform to the requirements of 4.5.2.3.3. Position the mount on the elevation adapter. Orient the mount as specified in 3.9d through 3.9g. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Insert the mandrel portion of the mounting bracket, with dioptometer, into the telescope clamping mechanism of the mount. The eyepiece of the dioptometer shall be on the side of the mount that contains the elevation knob. Rotate the mounting bracket, in the clamping mechanism, until the dioptometer vertical reticle line is plumb. Lock the clamping mechanism. Position the wall target at its calibrated distance. Focus the dioptometer on the wall target by means of the objective assembly. Elevate the target until the intersecting point of the horizontal and vertical target line is coincident to the intersecting point of the dioptometer horizontal and vertical reticle lines. Plumb the vertical target line independent of the dioptometer vertical retcile line.

Note: The condition of 3.9d shall be established with backlash removed from the elevation mechanism for rotation in depression.

4.6.5.1 Elevation travel. With backlash removed for direction of travel, and while directing a line of sight through the center of the dioptometer reticle pattern to the vertical target line, depress the line of sight of the mount in 200 mil increments by means of its elevation knob. Return the line of sight, through the dioptometer, to the horizontal plane by elevating the elevation and cross leveling fixture. Repeat the above steps until the mount is at maximum depression. Position the inspection aid (8658940) on the gunner's quadrant seats provided on the elevation adapter and verify the mil by measuring the angle established on the elevation adapter. Return the mount and elevation and cross leveling fixture to zero mil elevation. By means of the elevation knob, elevate the line of sight of the mount to its maximum elevation position. Again, by means of the Inspection Aid (8658940) verify the mil value indicated on the elevation scale and micrometer knob of the mount. The mount shall conform to the requirements of 3.10.

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4.6.5.2 Plumb travel. Perform this test concurrent with the test in 4.6.5.1. During the excursions in depression and elevation, the vertical reticle line of the dioptometer, at the center of the reticle pattern, shall track the vertical target line within the tolerance specified in 3.11, as measured on the target. Caution shall be exercised to assure that this test is conducted with backlash removed for direction of travel.

4.6.5.3 Lateral displacement. Perform this test in sequence to the test in 4.6.5.1 and 4.6.5.2. With backlash removed for direction of travel and the dioptometer reticle pattern superimposed on the vertical target line as specified in 4.6.6.1, track the vertical target line for approximately 50 mils by means of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob for approximately 50 mils. Any lateral displacement of the dioptopmeter vertical reticle line, as a result of this operation, shall not exceed the tolerance specified in 3.12 as measured on the target. Tests for lateral displacement shall be performed within 50 mils of each extremity of the elevation scale. A third test point for lateral displacement shall be performed at approximately mid-range of the elevation scale. All points of tests for lateral displacement shall be randomly selected within the above specified areas.

4.6.5.4 Elevation angular accuracy. Perform this test at 700 mils, 1000 mils, 1200 mils, 1400 mils, and 1600 mils. With backlash removed from fixture rotation in elevation, elevate the adapter to zero mil in elevation. The zero mil elevation position of the elevation adapter shall be verified with the Inspection Aid, 8658940. With backlash removed for rotation of a line of sight in depression, verify that the dioptometer reticle lines are superimposed on the target lines, and that the mount conforms to the orientation requirements of 3.9d through 3.9g. Depress the line of sight of the mount, by means of its elevation knob, to the first randomly selected elevation setting. Then by means of the elevation mechanism of the test fixture, return the line of sight of the mount to the horizontal plane until the dioptometer reticle lines are again superimposed on the target lines without overtravel. Utilizing the inspection aid, measure the elevation angle on the adapter. The measured angle recorded on the inspection aid must equal the angle indicated on the scale and micrometer of the mount within the tolerance specified in 3.15.1. Utilizing the foregoing method, proceed to inspect the elevation accuracy of the mount at the three remaining selected elevation settings. The one elevation setting below 800 mils shall be inspected from a zero mil setting of the fixture and a 800 mil setting of the mount. Record the elevation error for each elevation setting tested for use in the elevation backlash test.

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Caution must be exercised throughout all elevation accuracy measurements to exclude backlash from the mount and test fixture for their appropriate directions of travel. The elevation accuracy of the mount shall conform to the requirements of 3.15.1 and meet the requirements of 3.9d.

4.6.5.5 Elevation backlash. Perform this test in sequence to the test in 4.6.5.4. At each of the selected elevation settings, record the elevation (mil) value when the reticle lines are coincident with the target lines. With backlash excluded for direction of travel, override the target line (horizontal) by approximately 50 mils utilizing the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob and reestablish coincidence of the reticle and target lines (horizontal), without overtravel. Again record the elevation (mil) value as read on the scale and micrometer knob. The difference between the first and second recorded (mil) value is the amount of backlash. At the selected elevation setting, the amount of backlash for that elevation position shall not exceed the tolerance specified in 3.16.1. Perform the elevation backlash test at each of the selected elevation settings tested in 4.6.5.4.

4.6.6 Orientation, azimuth accuracy. The testing equipment utilized for these tests shall conform to the requirements of 4.5.2.3.4. Position the mount on the azimuth adapter. Adjust and position the dioptometer and mounting bracket on the mount as specified in 4.6.5. Orient the mount as specified in 3.9d through 3.9g. Verify that the vertical and horizontal reticle lines of the dioptometer and collimator target lines are superimposed. Then ascertain that the optical axis of the dioptometer is horizontal by rotating the wheel assembly of the fixture to 1600 mils, then counter rotating the mount, by means of the azimuth knob, to return the dioptometer vertical reticle line coincident to the collimator vertical target line. Correct any elevation error, as evidenced by the lack of coincidence of the horizontal reticle and collimator target lines, by adjusting the clamping mechanism for elevation of the line of sight. Then rotate the wheel assembly of the fixture to 4800 mils and repeat the above procedure to level the dioptometer. Repeat the entire procedure until the dioptometer is horizontal at 1600 mils, zero mils, and 4800 mils without interim adjustments. Return the fixture to the zero mil azimuth setting and the dioptometer reticle coincident to the target collimator vertical and horizontal lines.

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4.6.6.1 Horizontal travel. Verify that the intersecting point of the dioptometer reticle lines is coincident with the horizontal target line. With backlash removed for direction of travel, rotate the line of sight of the mount through 6400 mils, in precise 800 mil increments by means of the azimuth knob. At each 800 mil incremental position of the mount, counter-rotate the wheel assembly of the fixture 800 mils to reestablish coincidence of the dioptometer vertical reticle line and vertical target line. At each incremental position, note any lack of coincidence of the horizontal reticle line and horizontal target line. Any vertical displacement of the line of sight through the intersecting point of the dioptometer reticle lines, with reference to the horizontal target line, shall not exceed the tolerance specified in 3.14 as measured on the target collimator and meet the requirements of para 3.9d.

Note: Care shall be exercised to assure that backlash is excluded in all horizontal travel measurements. The 800 mil incremental poisitions of the mount and test fixture shall be accomplished without overtravel.

4.6.6.2 Vertical displacement. This test may be conducted concurrent with the test in 4.6.6.1. At each 800 mil incremental position, reverse the azimuth motion of the mount by means of the azimuth knob a minimum of 50 mils. As a result of this reversing operation, the line of sight, through the intersection of the dioptometer reticle lines, shall not displace vertically in excess of the tolerance specified in 3.13, as measured on the target collimator.

4.6.6.3 Azimuth angular accuracy. Perform this test at 800 mil incremental positions (8) as the line of sight of the mount is rotated 6400 mils in azimuth. Verify that the fixture wheel assembly is positioned at zero mil azimuth with backlash removed for appropriate movement in azimuth. With backlash removed for rotation of the line of sight of the mount to 800 mils, verify that the dioptometer vertical reticle line is superimposed on the vertical target line, and that the mount conforms to the requirements of 3.9d through 3.9g. Rotate the line of sight of the mount to 800 mils without overtravel. Counter rotate the fixture wheel assembly and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Any difference between the mil value recorded on the mount and the scale of the fixture is azimuth accuracy error. Record any azimuth accuracy error for use in the azimuth backlash test. Utilizing the foregoing method, proceed to inspect the azimuth accuracy of the mount at the seven (7) remaining incremental positions. Record the azimuth error for each of the eight (8) incremental positions tested. Caution must be exercised throughout all azimuth accuracy measurements to exclude backlash from the mount and test fixture for their appropriate direction of travel. The azimuth accuracy of the mount shall conform to the requirements of 3.15.2 and meet the requirements of para 3.9d.

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4.6.6.4 Azimuth backlash. Perform this test in sequence to the test in 4.6.6.3, at each of the 8 incremental positions. Subsequent to the azimuth angular accuracy measurements rotate the mount, by means of the azimuth knob, to an azimuth setting of approximately 850 mils. Then, with backlash included, reverse direction of the knob and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Record the azimuth mil value as read on the azimuth knob. Any azimuth change from the original 800 mil setting, as read on the azimuth knob scale, is the amount of backlash. At the selected azimuth position, the amount of backlash for that position added to the azimuth accuracy error recorded for that position, shall not exceed the tolerance specified in 3.16.2.

4.6.7 Operability. The following test shall be conducted with the mount positioned on a vibration free surface that has a mating dovetail that maintains the mount as specified in 3.9a and 3.9b. A mandrel that meets the physical requirements specified in 3.17.1 and torque testing equipment conforming to 4.5.2.3.5 shall also be utilized.

4.6.7.1 Telescope clamping mechanism, unlocked position. When subjected to a tactile examination, the function of the locking lever shall meet the requirements of 3.17.1.1. When the mandrel is inserted into the telescope holder, it shall rotate freely for 360 degrees and the telescope holder move fore and aft freely.

4.6.7.2 Telescope clamping mechanism, locked position. Insert the mandrel into the telescope holder. Apply a force of 15 to 17 inch-pounds to the locking lever and 10 to 12 inch-pounds to the wingnut. Apply the specified torques to the mandrel and telescope holder, in turn. The telescope clamping mechanism shall meet the requirements of 3.17.1.2.

4.6.7.3 Mount locking latch. Install and remove the mount from the dovetail on the stand a minimum of 3 times (3 installations). The mount shall meet the requirements of 3.17.2, at the temperature specified.

4.6.7.4 Locking knobs. Rotate the azimuth and elevation locking knobs to a locked position with the appropriate torque wrench and adapter. The azimuth and elevation mechanisms shall meet the requirements of 3.17.3 when the specified torque is applied to the applicable knob, at the temperature indicated.

4.6.7.5 Azimuth slip scales. When subjected to a tactile examination for operability, the micrometer knob slip scales (azimuth and elevation) and azimuth scale shall meet the requirements of 3.17.4, at the temperature indicated.

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4.6.7.6 Torque, azimuth, and elevation knobs. Utilizing the appropriate torque wrench with adapter, the running torque required to rotate the knobs throughout their range of rotation shall be within the values specified in 3.17.5, for the temperature indicated.

4.6.8 Illumination. The illumination test shall be performed under simulated conditions specified in 3.18. The luminosity test shall be performed in a dark room having a variably controlled light source. To perform this test, the observer must be dark adapted by being in a darkened room for a minimum of 15 minutes before viewing scale markings, indices, level vial bubbles, and markings. All scale markings, indices, level vial bubbles, and markings shall be clearly distinguishable when viewed, at a distance of 18 to 20 inches from the viewer's eye, with and without a 50% neutral density filter while the light level is varied gradually from zero to ten foot-candles over a period of one to two minutes.

4.6.9 Workmanship. The mount shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The mount shall conform to the requirements of 3.7.

5. PACKAGING

5.1 Preparation of radioactive material. Handling, packaging, packing, shipping, and marking of the Mount, Telescope 9360169 shall be in accordance with Packaging Data Sheet 9360169 and DSAM 4145.8. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Mount, Telescope: 9360169 is a component of Sightunit: M64A1. The Mount introduces measured azimuth and elevation readings to Elbow Telescope: 9360257. The Sightunit is used with the US Army mortar systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation.

c. Applicable stock number.

d. Applicable Packaging Data Sheet Number (See 5.1).

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e. Provisions for First Article Testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-10054. See 6.3.1.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001. When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-70453	MOUNT, TELESCOPE: 9360169

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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